Java Program

```
import java.util.Scanner;

class Calculator {
    Scanner sc;

    // Method to initialize Scanner
    void initialize() {
        sc = new Scanner(System.in);
    }

    // Method to display menu
    void displayMenu() {
        System.out.println("\n===== JAVA CALCULATOR =====");
        System.out.println("1. Addition");
        System.out.println("2. Subtraction");
        System.out.println("3. Multiplication");
        System.out.println("4. Division");
        System.out.println("5. Exit");
    }

    // Method to accept user's choice
    int acceptChoice() {
        while (true) {
            try {
                System.out.print("Enter your choice: ");
                int choice = Integer.valueOf(sc.nextLine());

                if (choice >= 1 && choice <= 5) {
                    return choice;
                } else {
                    System.out.println(
                        "Invalid choice! Enter 1 to 5."
                    );
                }
            } catch (NumberFormatException e) {
                System.out.println(
                    "Error! Enter a valid integer."
                );
            }
        }
    }

    // Method to accept a number
    double acceptNumber(String message) {
```

```
    while (true) {
        try {
            System.out.print(message);
            return Double.parseDouble(sc.nextLine());
        } catch (NumberFormatException e) {
            System.out.println(
                "Invalid input! Enter a valid number."
            );
        }
    }
}

// Method to perform calculations
void calculate(int choice, double a, double b) {
    double result;

    try {
        switch (choice) {
            case 1:
                result = a + b;
                System.out.println("Result = " + result);
                break;

            case 2:
                result = a - b;
                System.out.println("Result = " + result);
                break;

            case 3:
                result = a * b;
                System.out.println("Result = " + result);
                break;

            case 4:
                if (b == 0) {
                    throw new ArithmeticException(
                        "Division by zero is not allowed."
                    );
                }

                result = a / b;
                System.out.println("Result = " + result);
                break;

            default:
                System.out.println("Invalid operation!");
```

```
            }
        } catch (ArithmeticException e) {
            System.out.println("Error: " + e.getMessage());
        } finally {
            System.out.println("Operation completed.");
        }
    }

    // Method to run calculator repeatedly
    void runCalculator() {
        boolean running = true;

        while (running) {
            displayMenu();
            int choice = acceptChoice();

            if (choice == 5) {
                running = false;
                System.out.println(
                    "Calculator closed. Thank you!"
                );
            } else {
                double a = acceptNumber(
                    "Enter the first number: "
                );

                double b = acceptNumber(
                    "Enter the second number: "
                );

                calculate(choice, a, b);
            }
        }
    }
}
// Separate class containing main method
class Main {
    public static void main(String[] args) {
        Calculator obj = new Calculator();

        obj.initialize();
        obj.runCalculator();

        obj.sc.close();
    }
}
```

Sample Output

```
===== JAVA CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 1
Enter the first number: 25
Enter the second number: 15
Result = 40.0
Operation completed.

===== JAVA CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 2
Enter the first number: 25
Enter the second number: 15
Result = 10.0
Operation completed.

===== JAVA CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 3
Enter the first number: 2
Enter the second number: 5
Result = 10.0
Operation completed.
```

===== JAVA CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Enter your choice: 4
Enter the first number: 225
Enter the second number: 15
Result = 15.0
Operation completed.

===== JAVA CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Enter your choice: 4
Enter the first number: 225
Enter the second number: 0
ERROR!
Error: Division by zero is not allowed.
Operation completed.

===== JAVA CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Enter your choice: 5
Calculator closed. Thank you!